ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** B gets hotter than A	Thin wire has higher resistance.
3	**(a)** the iron rod	Iron concentrates the field.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Series connection allows it to sense all current.
5	**(c)** glows dimmer	Reversed cell opposes the others.
6	**(b)** the current is larger	More current.
7	**(c)** A is true but R is false.	A is true; R is false because a thick wire has less resistance.
8	**(a)** strength rises with turns	Doubling turns gives about double pins.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's observation is explained by the magnetic field of the current.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the loss of magnetism on switching off does not explain why the magnet is stronger.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Both feet are at the same potential (1); No current passes through the bird (1).	1+1
12	Total resistance increases (1); Less current for each (1).	1+1
13	Current decreases (1); The electromagnet becomes weaker (1).	1+1
14	10 A fuse (1); It is just above 9 A, so it protects without blowing in normal use (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1).	1+1+1
16	Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1).	1+1+1
17	The switch breaks the circuit (1); No current flows (1); But switching off the plug is also advised (1).	1+1+1
	OR: The heater (1); It carries a larger current (1); A thick wire has low resistance and does not overheat (1).	1+1+1
18	Both give deflection since current flows (1); Iron foil may change the field slightly (1); Paper has no effect (1).	1+1+1
	OR: Most energy goes to light, not heat (1); Filament is small (1); Not efficient (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) So that current flows through the turns and not directly between neighbouring turns/nail (1). (ii) More turns give a stronger electromagnet (8 to 17 pins) (1). (iii) It loses its magnetism (soft iron is a temporary magnet) (1). (iv) Use more cells (more current) / use a larger soft iron core (any two sensible) (2).	1+1+1+2
20	Total = 4.5 + 0.5 + 0.3 (1); = 5.3 A (1); Exceeds 5 A, so unsuitable (1); Fuse would melt in use (1); Choose a 10 A fuse for safety (1).	1+1+1+1+1
	OR: Risk of shock (1); Switch off at the main (1); Remove the plug (1); Use insulated tools (1); Test before touching (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.

- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** melt	Total 7 A exceeds 5 A.
2	**(c)** doubled	Heat is proportional to time.
3	**(c)** A is true but R is false.	A is true; R is false: it has a very high melting point.
4	**(c)** deflects from its north-south position	Current produces a magnetic field that acts on the needle.
5	**(a)** an electric shock	Live wire exposed.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Reason explains it.
7	**(b)** no current passes through the bird since there is no potential difference across its feet	No complete path.
8	**(d)** find and remove the cause such as overloading	Safety.
9	**(d)** A is false but R is true.	A fuse wire has a low melting point, so A is false; R is true.
10	**(c)** A is true but R is false.	A is true (heating effect); R is false, the filament becomes very hot.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No, only a coil produces a strong field (1); The nail needs a coil of wire carrying current (1).	1+1
12	A long wire fits in less space as a coil (1); Compact design gives high resistance and heat in a small area (1).	1+1
13	It melts first on high current (1); Breaks the circuit before wires or appliances are damaged (1).	1+1
14	Energy is wasted as heat (1); Use thick wire of low resistance or high voltage transmission (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Live meets neutral so resistance falls (1); Very large current heats wires (1); Fuse melts; MCB trips (1).	1+1+1
16	Wire has resistance so heat is produced (1); Any moving charge makes a magnetic field (1); Both effects occur together (1).	1+1+1
17	Thin wire has high resistance (1); More heat is made per second (1); Thick wires of low resistance are used for connection (1).	1+1+1
	OR: Old cell supplies less current (1); Weaker magnetic field (1); Strength depends on current (1).	1+1+1
18	Count pins lifted (1); Change one factor at a time (1); Plot a graph (1).	1+1+1
	OR: Coil: carries current (1); Core: gets magnetised (1); Armature, hammer and gong: strike and sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A safety device with a thin low-melting wire that breaks the circuit when current is too high (1). (ii) 10 A (1). (iii) Heating effect of current (1). (iv) A 30 A fuse would not melt at a current well above 8 A, so wires could overheat and cause a fire before the circuit breaks (2). A 5 A fuse would blow in normal use.	1+1+1+2
20	Resistance of wire converts electric energy to heat (1); Thin wire has more resistance (1); Overheating damages insulation (1); Use thicker wire for heavy loads (1); Fuses limit current (1).	1+1+1+1+1
	OR: In series same current (1); A has higher resistance, hotter (1); In parallel B draws a larger current (1); B still stays cooler due to low resistance (1); Heat depends on resistance and current (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.

- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** heating and magnetic	Kettle heats; bell uses magnetism.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** continues to heat	Independent paths.
4	**(c)** so leakage current flows to the ground and not through the user	Safety measure.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Excess current causes fire.
6	**(d)** 15 A	Just above the load.
7	**(a)** in circles around the wire	Circular.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R only describes the material, not why more turns strengthen the magnet.
10	**(a)** Both A and R are true and R is the correct explanation of A.	These properties make it suitable for heating elements.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A coin does not melt at a low current (1); Appliance may overheat and cause a fire (1).	1+1
12	Soft iron demagnetises quickly (1); The hammer can reset quickly (1).	1+1
13	The third pin is connected to earth (1); It carries leakage current safely away (1).	1+1
14	(a) Heating effect (1); (b) and (c) Magnetic effect (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It connects earth first when plugging (1); Disconnects last on removal (1); Ensures safety (1).	1+1+1
16	Soft iron loses magnetism at once (1); Steel stays magnetised (1); Release must be quick (1).	1+1+1
17	Magnetic only when current flows (1); Strength changes with current and turns (1); Poles can be reversed (1).	1+1+1
	OR: To cut off the live supply (1); A fuse on neutral leaves the appliance live (1); Hence a danger of shock (1).	1+1+1
18	Both are electric currents (1); Cell gives a small voltage (about 1.5 V) (1); Mains is about 220 V and dangerous (1).	1+1+1
	OR: Use correct rating (1); Switch off before replacing (1); Never use wire or coin (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hans Christian Oersted (1). (ii) Magnetic effect of electric current (1). (iii) No current, so no deflection, the needle stays pointing north (1). (iv) The direction of the magnetic field depends on the direction of the current: reversing the current reverses the deflection (2).	1+1+1+2
20	More turns (1); More current (1); Bigger soft iron core (1); If current fails the load falls (1); Safety zone and backup systems (1).	1+1+1+1+1
	OR: Overloading (1); Short circuit (1); Faulty wiring (1); Use proper rated fuses/MCB (1); Regular inspection and no sharing of sockets (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.

- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

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